

Research Interests	Databases, Systems, Cloud-Native Databases, Workload-Adaptive Data Systems	
Education	Massachusetts Institute of Technology	2020 – 2026
	Doctor of Philosophy in Computer Science (PhD)	Cambridge, MA
	• <i>Advisor:</i> Tim Kraska, Data Systems Group. • <i>Dissertation:</i> Virtualizing Data Infrastructures.	
	University of Toronto	2018 – 2020
	Master of Science in Computer Science (MSc)	Toronto, ON, Canada
	• <i>Advisor:</i> Gennady Pekhimenko. • <i>Research:</i> GPU performance prediction for deep neural network training.	
	University of Waterloo	2013 – 2018
	Bachelor of Software Engineering (BSE)	Waterloo, ON, Canada
Academic Appointments	Postdoctoral Associate	September 2026 – Present
	Massachusetts Institute of Technology	Cambridge, MA
	• <i>Host:</i> Tim Kraska, Computer Science and Artificial Intelligence Laboratory (CSAIL).	
Publications	Blueprinting the Cloud: Unifying and Automatically Optimizing Cloud Data Infrastructures with BRAD	
	Geoffrey X. Yu , Ziniu Wu, Ferdi Kossmann, Tianyu Li, Markos Markakis, Amadou Ngom, Samuel Madden, Tim Kraska	
	Proceedings of the VLDB Endowment (VLDB), Vol. 17, No. 11., 2024.	
	Check Out the Big Brain on BRAD: Simplifying Cloud Data Processing with Learned Automated Data Meshes	
	Tim Kraska*, Tianyu Li*, Samuel Madden*, Markos Markakis*, Amadou Ngom*, Ziniu Wu*, Geoffrey X. Yu *	
	Proceedings of the VLDB Endowment (VLDB), Vol. 16, No. 11, 2023. Vision Paper.	
	TreeLine: An Update-In-Place Key-Value Store for Modern Storage	
	Geoffrey X. Yu *, Markos Markakis*, Andreas Kipf*, Per-Åke Larson, Umar Farooq Minhas, Tim Kraska	
	Proceedings of the VLDB Endowment (VLDB), Vol. 16, No. 1, 2022.	
	Habitat: A Runtime-Based Computational Performance Predictor for Deep Neural Network Training	
	Geoffrey X. Yu , Yubo Gao, Pavel Golikov, Gennady Pekhimenko	
	USENIX Annual Technical Conference (USENIX ATC), 2021.	
	Skyline: Interactive In-Editor Computational Performance Profiling for Deep Neural Network Training	
	Geoffrey X. Yu , Tovi Grossman, Gennady Pekhimenko	
	ACM Symposium on User Interface Software and Technology (UIST), 2020.	
	* Denotes equal contribution.	
Preprints and Under Submission	EXPLAIN Yourself! Finding Query Planner Stalls Across DBMSes	
	Geoffrey X. Yu , Ryan Marcus, Tim Kraska	
	Under submission to the Conference on Innovative Data Systems Research (CIDR), 2027. arXiv Preprint arXiv:2608.23402 , 2026.	
	Tailwind: A Practical Framework for Query Accelerators	
	Geoffrey X. Yu , Ryan Marcus, Tim Kraska	
	arXiv Preprint arXiv:2604.28079 , 2026.	

JetStream: Generating Query Accelerators for Existing Database Systems

Akhilesh Balasingam, Amadou Ngom, [Geoffrey X. Yu](#), Tim Kraska
arXiv Preprint [arXiv:2608.22476](#), 2026.

Demonstrations

Virtualizing Cloud Data Infrastructures with BRAD

[Geoffrey X. Yu](#), Ziniu Wu, Ferdi Kossmann, Tianyu Li, Markos Markakis, Amadou Ngom, Sophie Zhang, Tim Kraska, Samuel Madden
International Conference on Management of Data (SIGMOD), 2025.

TBD Suite: Benchmarking and Profiling Tools for DNNs

[Geoffrey X. Yu](#), Hongyu Zhu, Anand Jayarajan, Bojian Zheng, Abhishek Tiwari, Gennady Pekhimenko
Conference on Machine Learning and Systems (MLSys), 2019.

Professional Experience

Amazon Web Services

Boston, MA

Applied Scientist Intern

May – August 2024

- *Mentor:* [Vikram Nathan](#).
- Led research on designing and simulating strategies for per-query configurable per-query configurable cost-performance trade-offs in Redshift Serverless' intelligent scaling mode.

Intel Corporation

Remote

Research Graduate Intern

May – August 2021

- *Mentors:* [Nesime Tatbul](#) and David Cohen.
- Led research on new designs for concurrent learned indexes that support database transactions.

Meta Platforms (formerly Facebook)

Menlo Park, CA

Software Engineering Intern

January – April & September – December 2017

Apple

Sunnyvale, CA

Software Engineering Intern

May – August 2016

- [Contributed to Apache Cassandra](#) to improve observability and data repair capabilities.

Coursera

Mountain View, CA

Software Engineering Intern

January – April & September – December 2015

Teaching Experience

Teaching Assistant

6.5830 / 6.5831 – Database Systems

Massachusetts Institute of Technology

Fall 2022

- *Instructors:* [Samuel Madden](#) and [Michael Cafarella](#)
- *Overall Rating:* 6.7 out of 7 ($\sigma = 0.48$, $n = 10$)

Teaching Assistant

CSC 209 – Software Tools and Systems Programming

University of Toronto

Fall 2018, Winter 2019

Student Mentoring

- **Sophie Zhang**
Undergraduate research, Master's thesis (MEng)
↔ Next: Software Engineer at Databricks

Massachusetts Institute of Technology

2024 – 2025

- **Yubo Gao**
Undergraduate research
↔ Next: PhD student at the University of Toronto

University of Toronto

2019 – 2020

- **Pavel Golikov**
Undergraduate research
↔ Next: PhD student at the University of Toronto

University of Toronto

2019 – 2020

Service

- VLDB 2026 Demonstration Track Program Committee

Awards and Honors

- **NSERC Postgraduate Scholarship – Doctoral (PGS D)** 2020 – 2023
Merit-based doctoral research scholarship awarded by the Government of Canada (63,000 CAD).
- **NSERC Alexander Graham Bell Canada Graduate Scholarship (CGS D)** 2020 – 2023 (declined)
Merit-based doctoral research scholarship awarded by the Government of Canada (105,000 CAD).
- **NSERC Canada Graduate Scholarship – Master’s (CGS M)** 2019 – 2020
Merit-based master’s research scholarship awarded by the Government of Canada (17,500 CAD).
- **Snap Research Scholarship** January 2019
Awarded by Snap Inc. for research quality, technical skills, and communication skills (10,000 USD).
- **Vector Institute Scholarship in Artificial Intelligence** 2018 – 2019
Awarded to top students pursuing AI-related master’s studies in Ontario (17,500 CAD).
- **Queen Elizabeth II Graduate Scholarship in Science and Technology** 2018 – 2019
Merit-based scholarship awarded to students pursuing graduate studies in Ontario (15,000 CAD).
- **Wolfond Fellowship, University of Toronto** 2018 – 2019
Awarded at the University of Toronto for academic achievements (10,000 CAD).
- **Sandford Fleming Foundation Award for Academic Excellence** June 2018
Best graduating academic record in the Software Engineering program at the University of Waterloo.

Invited Talks

- **Tailwind: A Practical Framework for Query Accelerators** May 21, 2026
InterSystems Corporation Boston, MA
- **Virtualizing Cloud Data Infrastructures with BRAD** May 23, 2024
North East Database Day 2024 Boston, MA
- **Check Out the Big Brain on BRAD: Simplifying Cloud Data Processing with Learned Automated Data Meshes** September 15, 2023
Cockroach Labs Virtual
- **Learning-Based Creation of Data Mesh Architectures** March 10, 2023
North East Database Day 2023 Boston, MA